CFGs

not cover: • recursive FA / NFA
pushdown FA

• prove that language is not context-free

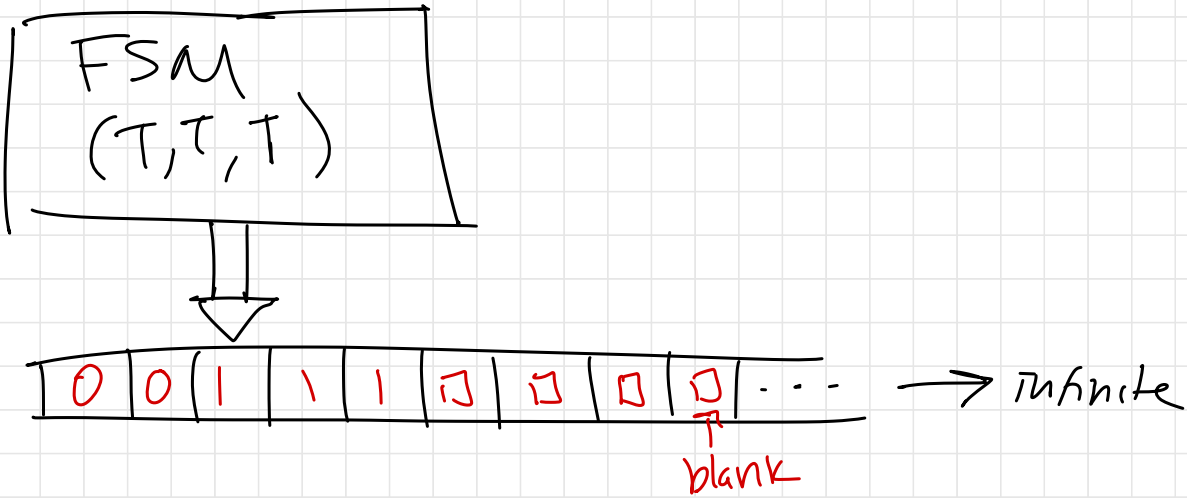
$\{0^n 1^n 0^n : n \geq 0\}$

$\{ww : w \in \Sigma^*\}$

• closure properties

capability	language	machine
sequencing branching repetition	regular	DFA / NFA
recursion	context-free language	recursive FA
(infinite) memory	decidable	"arbitrary computation" → python → RISC Turing machine

Turing Machines



At each step:

- reads symbol from tape a
- based on symbol and state of FSM:
  - write new symbol to tape
  - change state
  - move L or R

δ

ex TM that recognizes $\{0^n 1^n 0^n : n \geq 0\}$

$TM = (\Gamma, \square, \Sigma, Q, q_{start}, q_{reject}, q_{accept}, \delta)$

Γ = tape alphabet

Gamma

$\square \in \Gamma$ blank symbol

$\Sigma \subseteq (\Gamma \setminus \square)$ = input alphabet

sigma

Q = states

$q_{start}, q_{reject}, q_{accept} \in Q$

once a TM enters q_{reject} or q_{accept}, it halts

$\delta(\Gamma \times Q \setminus \{q_{reject}, q_{accept}\}) \rightarrow$

$Q \times \Gamma \times \{+1, -1\}$

move left move right

010

110011 X

$\Gamma = \{0, 1, \$, x, \square\}$

$\Sigma = \{0, 1\}$

$Q = \{start, seek1, seek0, reset, verify, accept, reject\}$

$	0	x	1	x	0	□	□	...

start - mark 0 w/ a $

seek1 - look for + mark corresponding 1 with an x

seek0 - look for corresponding 0 + mark with an x

reset - go back to first group of 0s

verify - sweep over string for unmatched chars.

$\delta(\ p\ , a) = (\ q\ , b, \delta\ )$	explanation
$\delta($ start , 0$) = ($ seek1 , \$, +1$)$	mark first 0 and scan right
$\delta($ start , x$) = ($ verify , \$, +1$)$	looks like we're done, but let's make sure
$\delta($seek1, 0$) = ($ seek1 , 0, +1$)$	scan rightward for 1
$\delta($seek1, x$) = ($ seek1 , x, +1$)$	
$\delta($seek1, 1$) = ($seek0 , x, +1$)$	mark 1 and continue right
$\delta($seek0, 1$) = ($seek0 , 1, +1$)$	scan rightward for 0
$\delta($seek0, x$) = ($seek0 , x, +1$)$	
$\delta($seek0, 0$) = ($ reset , x, +1$)$	mark 0 and scan left
$\delta($ reset , 0$) = ($ reset , 0, −1$)$	scan leftward for \$
$\delta($ reset , 1$) = ($ reset , 1, −1$)$	
$\delta($ reset , x$) = ($ reset , x, −1$)$	
$\delta($ reset , \$$) = ($ start , \$, +1$)$	step right and start over
$\delta($verify, x$) = ($ verify , \$, +1$)$	scan right for any unmarked symbol
$\delta($verify, □$) = ($accept, □, −1$)$	success!

(start, 001100)
⇒ (seek1, \$01100)
⇒ (seek1, \$01100)
⇒ (seek0, \$0x100)
⇒ (seek0, \$0x100)
⇒ (reset, \$0x1x0)
⇒ (reset, \$0x1x0)
⇒ (reset, \$0x1x0)
⇒ (reset, \$0x1x0)
⇒ (start, \$0x1x0)
⇒ (seek1, \$\$x1x0)
⇒ (seek1, \$\$x1x0)
⇒ (seek0, \$\$xxx0)
⇒ (seek0, \$\$xxx0)
⇒ (reset, \$\$xxxx)
⇒ (reset, \$\$xxxx)
⇒ (reset, \$\$xxxx)
⇒ (reset, \$\$xxxx)
⇒ (start, \$\$xxxx)
⇒ (verify, \$\$\$xxx)
⇒ (verify, \$\$\$\$xx)
⇒ (verify, \$\$\$\$\$x)
⇒ (verify, \$\$\$\$\$\$□)
⇒ (accept, \$\$\$\$\$\$) ⇒ **accept!**

(start, 00100)
⇒ (seek1, \$0100)
⇒ (seek1, \$0100)
⇒ (seek0, \$0x00)
⇒ (reset, \$0xx0)
⇒ (reset, \$0xx0)
⇒ (reset, \$0xx0)
⇒ (start, \$0xx0)
⇒ (seek1, \$\$xx0)
⇒ (seek1, \$\$xx0)
⇒ (seek1, \$\$xx0) ⇒ **reject!**